Saturation-Aware Stable-Manifold Optimal Control of a Hysteretic Soft Pneumatic Actuator

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Abstract: We design a nonlinear optimal regulator for a soft pneumatic bending actuator with pressure lag, Bouc–Wen-type hysteresis and a bounded command. The command is integrated from an optimized rate, and its smooth saturation enters the drift before the Hamiltonian is formed, so the problem stays input-affine and the optimal rate is explicit. The stable manifold of the resulting five-state Hamiltonian system is computed by integral iteration, a positive-definite value function is fitted to manifold costates and Hamilton–Jacobi–Bellman collocation, and a guarded selector falls back to LQR. In simulation, regulation cost drops by 20.8% at the price of slower angle settling.

Keywords: Nonlinear optimal control; Hamilton–Jacobi–Bellman equation; stable manifold; input saturation; hysteresis; soft robotics; pneumatic actuators.

1. INTRODUCTION

Fluid-driven soft actuators are compliant and inherently safe, but their dynamics combine a pneumatic pressure lag, rate-dependent hysteresis of the elastomer and a command that is physically bounded by the valve and supply (Rus and Tolley, 2015; Polygerinos et al., 2017; Della Santina et al., 2023). Hysteresis in pneumatic actuators has been described by Maxwell-slip (Vo-Minh et al., 2011) and Prandtl–Ishlinskii operators (Xie et al., 2018) and compensated by inverse models (Aschemann and Schindele, 2014). The Bouc–Wen law (Wen, 1976; Ismail et al., 2009) instead gives a smooth differential model that can be placed directly inside a model-based design. Optimal feedback that treats hysteresis and the command bound together is still uncommon: model predictive control has been used for pneumatic soft robots (Best et al., 2016), and a hybrid Hamilton–Jacobi–Bellman (HJB) formulation was proposed for Preisach hysteresis in a magnetostrictive actuator (Tan and Baras, 2002).

The infinite-horizon optimal feedback solves a stationary HJB equation. Local power series (Al'brekht, 1961; Lukes, 1969), Galerkin projection (Beard et al., 1997), patchy expansions (Navasca and Krener, 2007) and algebraic dynamic solutions (Sassano and Astolfi, 2012) address it near an equilibrium, while characteristics-based data generation combined with regression scales to higher dimensions (Kang and Wilcox, 2017; Kalise and Kunisch, 2018; Azmi et al., 2021; Nakamura-Zimmerer et al., 2021). The stable manifold approach exploits the fact that the stabilizing HJB solution is generated by the stable Lagrangian submanifold of the associated Hamiltonian system (Sakamoto, 2002; Sakamoto and van der Schaft, 2008). This submanifold can be computed by an integral iteration along optimal trajectories. The approach has been validated experimentally (Sakamoto, 2013; Horibe and Sakamoto, 2018, 2019), extended with symplectic integrators (Chen and Zhu, 2022), and its existence conditions have been clarified (Sakamoto, 2023).

Input bounds are usually handled by non-quadratic input penalties, which yield a bounded minimizer but change the cost (Abu-Khalaf and Lewis, 2005), or by saturation functions combined with system extension (Graichen and Petit, 2009; Graichen et al., 2010). Constrained stable-manifold designs were developed by Tran and Sakamoto (2016) and Tran et al. (2017). In anti-windup terms (Galeani et al., 2009; Tarbouriech et al., 2011), an integrator state may leave the actuator range while the applied signal cannot.

This paper treats the bound as a property of the plant rather than of the cost. Its contributions are as follows. (i) A saturation-aware augmented formulation in which the applied command $\sigma(u)$ is a state nonlinearity of the drift. The design remains affine in the optimized rate, the rate minimizer is explicit, and the canonical equations inherit the saturation slope $\sigma'(u)$ by symbolic differentiation (Proposition 2). (ii) A five-state stable-manifold computation with explicit acceptance tests, together with a positive-definite scalar value approximation fitted jointly to manifold costates and HJB collocation and checked on held-out states. (iii) A guarded selector between the Hamilton–Jacobi (HJ) rate and a linear quadratic regulator (LQR) rate, and a simulation study that reports both the gains and the trade-offs, together with an analytic explanation of the slow hysteresis-relaxation mode that limits both controllers.

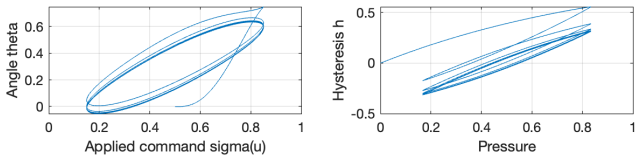


Fig. 1. Open-loop response to the applied command $\sigma(u) = 0.5{+}0.35\sin(\pi t/2)$ over 40 s, with weak forcing ($B_d = 0.15$): angle versus applied command (left) and hysteresis state versus pressure (right).

2. ACTUATOR MODEL AND AUGMENTATION

2.1 Pressure-driven hysteretic bending actuator

We consider a normalized single-segment bending actuator with angle θ (rad), angular velocity $v = \dot\theta$, normalized chamber pressure p and an internal hysteresis state h:

$$\dot\theta = v, \qquad J\dot v = -k\theta - cv + b_p p + k_h h + B_d d, \tag{1}$$

$$\dot p = r := \big(-p + k_u\sigma(u)\big)/\tau_p, \tag{2}$$

$$\dot h = \big(A_h - \gamma s_{\varepsilon_h}(h)\big)r - \beta h\, s_{\varepsilon_r}(r) - \lambda_h(h - h_r), \tag{3}$$

where $s_\varepsilon(a) = \sqrt{a^2+\varepsilon^2} - \varepsilon$ satisfies $s_\varepsilon(0) = 0$ and $0 \le |a| - s_\varepsilon(a) \le \varepsilon$. Equation (3) is the Bouc–Wen law with exponent one (Wen, 1976; Ismail et al., 2009), driven by the pressure rate r. The absolute values are replaced by the smooth surrogate s_ε, and a relaxation term returns h to h_r when the pressure is constant. The torque $B_d d$ is produced by an exosystem $\dot\eta = A_\eta\eta + B_\eta w$, $d = C_d\eta$, which is used only in simulation. The normalized demonstration parameters are $J = 0.02$, $k = 0.6$, $c = 0.08$, $b_p = 0.35$, $k_h = 0.25$, $\tau_p = 0.2\,\mathrm{s}$, $k_u = 1$, $A_h = 1$, $\gamma = 0.2$, $\beta = 0.8$, $\lambda_h = 0.02\,\mathrm{s}^{-1}$, $h_r = 0$ and $\varepsilon_h = \varepsilon_r = 10^{-4}$. They are representative values and were not identified on a specific device. Fig. 1 shows the open-loop response to a prescribed applied command.

2.2 Smooth saturation and rate augmentation

The applied command is bounded by the smooth saturation

$$\sigma(u) = u_m + u_h \tanh\big((u - u_m)/u_h\big), \tag{4}$$

with $u_m = u_h = 0.5$ and slope $\sigma'(u) = 1 - \tanh^2\big((u - u_m)/u_h\big)$. Hence $\sigma : \mathbb{R} \to (0,1)$ and $\sigma'(u) \in (0,1]$. The raw command u is generated by an integrator, $\dot u = \nu$, and the rate ν is the decision variable. With $z = [\theta, v, p, h, u]^\top$,

$$\dot z = \mathcal{F}_\sigma(z) + G\nu + Ed, \qquad G = e_5, \quad E = (B_d/J)\, e_2, \tag{5}$$

where e_i is the i-th unit vector and the drift $\mathcal{F}_\sigma$ contains $\sigma(u)$ through (2)–(3). For $\nu = 0$ and $d = 0$ the equilibrium satisfies $v^\star = 0$, $r^\star = 0$, $h^\star = h_r = 0$, $p^\star = k_u\sigma(u^\star)$ and $k\theta^\star = b_p p^\star$. The target $\theta^\star = 0.35$ gives $p^\star = 0.6$ and $u^\star = \sigma^{-1}(0.6) = u_m + u_h\,\mathrm{atanh}(0.2) = 0.6013662770$, at which $\sigma'(u^\star) = 0.96$. The resulting sampled loop is shown in Fig. 2.

Remark 1. If $\sigma(u)$ were the input, the plant would not be affine in the decision variable and the Hamiltonian minimizer would not be explicit. After augmentation, σ is a smooth function of the state and the pre-Hamiltonian is quadratic in ν. Unlike Graichen et al. (2010), where saturation functions are introduced as a design device, here σ models the actuator itself. The raw state u can leave

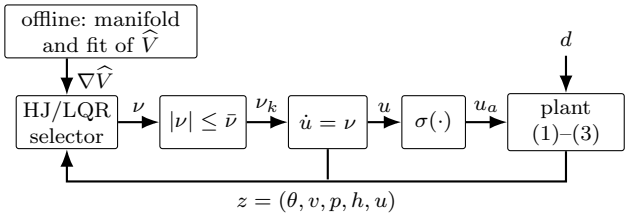


Fig. 2. Sampled loop: the selected raw rate is held, integrated to the raw command u, and saturated to the applied command $u_a = \sigma(u)$.

$[0,1]$, much like a wound-up integrator (Galeani et al., 2009), whereas $\sigma(u)$ cannot. A state penalty on $u - u^\star$ limits this drift.

3. SATURATION-AWARE HJB EQUATION AND STABLE MANIFOLD

3.1 Optimal control problem and Hamiltonian

With $e_z = z - z^\star$, we minimize

$$\mathcal{J} = \int_0^\infty \big(e_z^\top Q_z e_z + R_\nu \nu^2\big)\mathrm{d}t, \tag{6}$$

where $Q_z = \mathrm{diag}(80, 5, 2, 1, 0.2)$ and $R_\nu = 0.02$. The angle error dominates the cost, and the small weight $q_u = 0.2$ on the raw-command error limits its drift. For conditioning, $e_z = Sy$ with $S = \mathrm{diag}(0.35, 1, 0.6, 0.4, 0.6)$, whose entries are the typical sizes of the errors. This gives $Q_y = SQ_zS$, $a_\sigma(y) = S^{-1}\mathcal{F}_\sigma(z^\star + Sy)$, $G_y = S^{-1}G$ and the scaled costate $\mu = S\lambda$, where λ is the physical costate. The pre-Hamiltonian is

$$\mathcal{H}(y, \mu, \nu) = \mu^\top\big(a_\sigma(y) + G_y\nu\big) + y^\top Q_y y + R_\nu\nu^2. \tag{7}$$

Proposition 2. (a) $\mathcal{H}$ is strictly convex in ν. Its unique minimizer and the minimized Hamiltonian are

$$\nu^* = -\frac{G_y^\top\mu}{2R_\nu} = -\frac{\mu_5}{2R_\nu s_u}, \tag{8}$$

$$\overline{H}(y,\mu) = \mu^\top a_\sigma(y) + y^\top Q_y y - \frac{(G_y^\top\mu)^2}{4R_\nu}, \tag{9}$$

where $s_u = 0.6$ is the fifth scale. (b) Along the canonical flow of $\overline{H}$, the physical command costate obeys

$$\dot\lambda_5 = -\frac{k_u}{\tau_p}\sigma'(u)\big(\lambda_3 + D_r\lambda_4\big) - 2q_u(u - u^\star), \tag{10}$$

with $D_r = \partial\dot h/\partial r = A_h - \gamma s_{\varepsilon_h}(h) - \beta h\, r/\sqrt{r^2+\varepsilon_r^2}$. (c) For any measurable rate, $0 < \sigma(u(t)) < 1$ and $|\frac{\mathrm{d}}{\mathrm{d}t}\sigma(u(t))| \le |\nu(t)|$.

Proof. (a) $\partial^2\mathcal{H}/\partial\nu^2 = 2R_\nu > 0$; completing the square gives (8)–(9). (b) $\dot\lambda = -\partial\overline{H}/\partial e_z$, and u enters $\mathcal{F}_\sigma$ only through r in (2)–(3), with $\partial r/\partial u = k_u\sigma'(u)/\tau_p$. The state cost contributes $2q_u(u - u^\star)$. (c) This follows from the range of tanh, the chain rule and $0 < \sigma' \le 1$. □

The command costate is coupled to the pressure and hysteresis costates only through $\sigma'(u)$. Deep in saturation $\sigma' \to 0$, and (10) is driven by the penalty alone. At $u^\star$, $\sigma' = 0.96$, so even the linearized design differs from that of an unsaturated actuator. In the implementation, (7)–(10) are not entered by hand. They are generated symbolically by forming $\mathcal{H}$, solving $\partial_\nu\mathcal{H} = 0$, substituting, and differentiating $\overline{H}$.

3.2 HJB equation and its characteristics

The value function V solves the stationary HJB equation

$$\overline{H}\big(y, \nabla V(y)\big) = 0, \quad V(0) = 0, \qquad \nu_{\mathrm{HJ}}(y) = -\frac{G_y^\top \nabla V(y)}{2R_\nu}. \tag{11}$$

Its characteristics are the canonical equations

$$\dot{y} = a_\sigma(y) - \frac{G_y G_y^\top \mu}{2R_\nu}, \qquad \dot{\mu} = -\Big(\frac{\partial a_\sigma}{\partial y}\Big)^\top \mu - 2Q_y y. \tag{12}$$

Let $A_5 = \partial a_\sigma/\partial y(0)$, $B_5 = G_y$, let P_5 be the stabilizing solution of $A_5^\top P_5 + P_5 A_5 - P_5 B_5 R_\nu^{-1} B_5^\top P_5 + Q_y = 0$, and let $K_5 = R_\nu^{-1} B_5^\top P_5$, $\Gamma = 2P_5$, $D = B_5 B_5^\top/(2R_\nu)$ and $F_c = A_5 - B_5 K_5$. The linearization of (12) is the Hamiltonian matrix $\mathcal{A}_H = \left[\begin{smallmatrix} A_5 & -D \\ -2Q_y & -A_5^\top \end{smallmatrix}\right]$.

Lemma 3. (Sakamoto and van der Schaft 2008). Here $\mathcal{F}_\sigma$ is real-analytic (tanh and s_ε are analytic), (A_5, B_5) is controllable and $Q_y \succ 0$. Hence $\mathcal{A}_H$ is hyperbolic. Near the origin, the stable manifold of (12) is the Lagrangian graph $\{(y, \nabla V(y))\}$ of the stabilizing solution of (11), with $V(y) = y^\top P_5 y + O(\|y\|^3)$.

Let $L = L^\top$ solve $F_c L + L F_c^\top = D$ and $T = \left[\begin{smallmatrix} I & L \\ \Gamma & \Gamma L + I \end{smallmatrix}\right]$. The change of coordinates $(y, \mu) = T(s, \chi)$ gives $T^{-1}\mathcal{A}_H T = \mathrm{diag}(F_c, -F_c^\top)$ and

$$\dot{s} = F_c s + n_s(s, \chi), \qquad \dot{\chi} = -F_c^\top \chi + n_u(s, \chi),$$

where (n_s, n_u) collects the nonlinear remainder of (12). For a seed $\xi = s(0)$, a trajectory on the stable manifold is a fixed point of

$$\begin{aligned} s_{k+1}(t) &= e^{F_c t}\xi + \int_0^t e^{F_c(t-a)} n_s\big(s_k(a), \chi_k(a)\big)\mathrm{d}a, \\ \chi_{k+1}(t) &= -\int_t^{T_m} e^{F_c^\top(a-t)} n_u\big(s_k(a), \chi_k(a)\big)\mathrm{d}a, \end{aligned} \tag{13}$$

initialized with $s_0 = e^{F_c t}\xi$ and $\chi_0 = 0$ (Sakamoto and van der Schaft, 2008). Both kernels decay because F_c is Hurwitz. The initial state is $y(0) = \xi + L\chi(0)$, so a seed is not a prescribed initial state. We take $T_m = \max(40,\, 8/|\max \mathrm{Re}\,\lambda(F_c)|)$ and a non-uniform mesh (steps of 0.01 s on $[0, 4]$, 0.1 s on $[4, 30]$ and 0.5 s up to T_m). The forcing is interpolated linearly on each interval with exact exponential weights, and each update is relaxed by 0.8. The seeds are $\pm e_i$ and eight Halton directions, all of radius 2×10^{-4}. A trajectory is accepted only if the tests of Table 1 pass.

4. VALUE APPROXIMATION AND GUARDED FEEDBACK

4.1 Positive-definite value ansatz

The manifold supplies exact costates only near the origin, whereas the feedback must act from a cold start. We therefore fit one scalar function

$$\widehat{V}(y) = \big(y^\top P_5 y\big) e^{g(y)}, \qquad g(y) = \sum_{1 \le |\alpha| \le 3} c_\alpha y^\alpha, \tag{14}$$

with $5 + 15 + 35 = 55$ coefficients c_α, and $\nabla\widehat{V} = e^g\big[2P_5 y + (y^\top P_5 y)\nabla g\big]$.

Proposition 4. For every coefficient vector c, $\widehat{V}(0) = 0$, $\widehat{V}(y) > 0$ for $y \neq 0$, $\nabla\widehat{V}(0) = 0$ and $\nabla^2\widehat{V}(0) = 2P_5$. Hence $\widehat{V}$ reproduces the LQR value to second order, and the five costate components are the gradient of a single function.

Proof. $P_5 \succ 0$ and $e^g > 0$. Since $g(0) = 0$, $\widehat{V} = y^\top P_5 y + O(\|y\|^3)$. □

The coefficients are obtained from

$$\min_c \sum_i \big[w_i R_H(y_i)\big]^2 + \omega_a \sum_j \frac{\|\nabla\widehat{V}(y_j^m) - \mu_j^m\|^2}{a_j^2} + \lambda_c\|c\|^2, \tag{15}$$

where $R_H(y) = \overline{H}(y, \nabla\widehat{V}(y))$, (y_j^m, μ_j^m) are the manifold anchors, $a_j = \max(10^{-7}, \|\mu_j^m\|)$, $w_i = (0.03 + \ell_i)^{-1}$ with running cost $\ell = y^\top Q_y y + R_\nu \nu^{*2}$, $\omega_a = 0.05$ and $\lambda_c = 0.005$. The collocation set contains multi-scale Halton points (Halton, 1960), points on the coordinate axes and 250 states of the LQR closed loop, all restricted to $\|y\| \le 1.6$. We minimize (15) by Gauss–Newton steps with frozen weights and analytic Jacobians, using a backtracking line search that also enforces $|g| \le 3$. Like gradient-augmented regression (Azmi et al., 2021), the fit uses costate data, but here these data come from the stable manifold. HJB accuracy is measured on held-out states by

$$\rho_H = \frac{|R_H|}{10^{-10} + \ell + \big|\nabla\widehat{V}^\top(a_\sigma + G_y\nu^*)\big|} \in [0, 1], \tag{16}$$

which compares the residual with the two terms that must cancel.

4.2 Guarded rate selection

At each sample $t_k = kT_s$ the controller (Fig. 2) proceeds as follows. (i) It forms $y = S^{-1}(z - z^\star)$ with the raw u and the bounded LQR rate $\nu_L = \mathrm{sat}_{\bar\nu}(-K_5 y)$, where $\bar\nu = 20$. (ii) It declares HJ eligible if $\|y\| \le 1.6$, $\widehat{V}$ and $\nabla\widehat{V}$ are finite, $|g(y)| \le 3$ and $\rho_H(y) \le 0.05$. (iii) It computes $\nu_H = \mathrm{sat}_{\bar\nu}\big(-G_y^\top\nabla\widehat{V}/(2R_\nu)\big)$. Because $\mathcal{H}$ is a convex parabola in ν, ν_H is the exact minimizer of $\mathcal{H}$ over $[-\bar\nu, \bar\nu]$. (iv) It applies ν_H only if

$$\nabla\widehat{V}(y)^\top\big(a_\sigma(y) + G_y\nu_H\big) \le -0.02\, y^\top Q_y y, \tag{17}$$

and otherwise applies ν_L. The rate is held, and $u_{k+1} = u_k + T_s\nu_k$. For the exact value and the unconstrained rate, $\dot{V} = -\ell \le -y^\top Q_y y$, so (17) requires 2% of this ideal decrease along the nominal flow.

Remark 5. Inequality (17) is checked pointwise for $d = 0$ at sample instants. It does not certify the inter-sample behaviour, stability under switching between the two laws (Liberzon, 2003), or disturbance rejection. The rate bound is applied after an unconstrained-rate HJB design. It is not the solution of a rate-constrained HJB equation.

5. SIMULATION RESULTS

The loop runs with $T_s = 0.01$ s for 240 s (24 000 held intervals). Each sample uses two fourth-order Runge–Kutta substeps, and $\sigma(u)$ is evaluated at every stage. Both controllers start from rest, $z(0) = 0$. Then $\sigma(0) = 0.1192$ and $\|y(0)\| = 1.733 > 1.6$, so the first rates are LQR. The baseline is the same K_5 with the same rate bound. The finite-horizon cost $\mathcal{J}_{240} = \int_0^{240} e_z^\top Q_z e_z\,\mathrm{d}t + \sum_k T_s R_\nu \nu_k^2$

Table 1. Acceptance tests of the 18 manifold trajectories

Test	Threshold	Worst seed
Relative $\lvert\overline{H}\rvert$ along trajectory	5×10^{-3}	1.28×10^{-5}
Midpoint forcing defect	5×10^{-7}	2.13×10^{-7}
Fixed-point relative residual	10^{-7}	1.34×10^{-8}
Terminal norm $\lVert y(T_m)\rVert$	2×10^{-6}	7.24×10^{-8}
Boundary residual	10^{-10}	6.8×10^{-21}

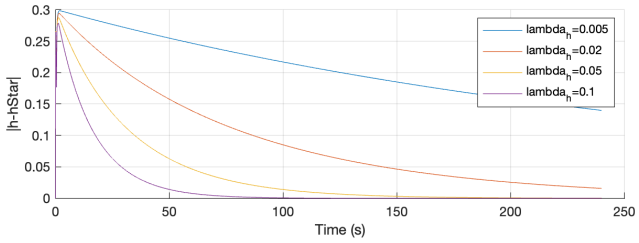


Fig. 3. Hysteresis offset $|h - h^\star|$ under the augmented LQR, retuned for each relaxation rate λ_h (s^{-1}), with $B_d = 0$.

splits into a state part and a rate part. The tail root-mean-square error (RMSE) of the angle is computed over the last 30 s, and the settling time is the instant after the last exit from $|\theta - \theta^\star| \le 0.007$ rad.

5.1 Local design and stable manifold

The poles of F_c are $-6.620 \pm 11.619\,\mathrm{i}$, -12.493, -4.158 and $-0.01190\,\mathrm{s}^{-1}$, which gives $T_m = 672.5$ s. The physical gain is $K_5 S^{-1} = [-1.674, 11.236, 17.590, 26.795, 20.883]$. The symbolic pipeline reproduces stationarity, substitution and the chain rule (10) to 1.1×10^{-16}, 7.1×10^{-15} and 2.2×10^{-16}, respectively. The Riccati and block-diagonalization residuals are 2.1×10^{-13} and 4.8×10^{-13}. All 18 seeds were accepted after 5–8 iterations (Table 1). They provide 1 050 anchors that span all five directions, with a largest radius of 2.63×10^{-4}.

Remark 6. The slowest closed-loop pole, $-0.0119\,\mathrm{s}^{-1}$, has a time constant of 84 s that can be explained analytically. Holding the angle at $\theta^\star$ with $v = 0$ requires $b_p p + k_h h = k\theta^\star$, so that $\dot{p} = -(k_h/b_p)\dot{h}$. Linearizing (3) at $h = r = 0$ gives $\dot{h} \approx A_h \dot{p} - \lambda_h h$, and hence

$$\dot{h} \approx -\frac{\lambda_h}{1 + A_h k_h / b_p}\, h = -0.01167\, h,$$

which lies within 2% of the computed pole. The hysteresis can therefore relax only at a fraction of λ_h: any faster decay would require pressure excursions that move the angle, and the angle carries the dominant weight. A sweep that retunes the LQR for each λ_h supports this formula (Fig. 3). Taking $h \approx 0.29$ after the initial transient ($t \approx 2$ s), the final offsets $h(240) = 0.140$, 0.0159, 2.2×10^{-4} and 1.8×10^{-7} for $\lambda_h = 0.005$, 0.02, 0.05 and $0.1\,\mathrm{s}^{-1}$ imply mean decay rates within 5% of the predicted 0.0029, 0.0117, 0.0292 and $0.0583\,\mathrm{s}^{-1}$.

5.2 Value fit

At the anchors, the relative gradient error has median 1.84×10^{-5}, 90th percentile 3.58×10^{-5} and maximum 1.03×10^{-4}. Nine Gauss–Newton steps reduce (15) from 7 363 to 284 for the 1 294 collocation states. On 500 held-out Halton states with $\lVert y\rVert \le 1.6$, the median, 90th and 99th percentiles of ρ_H are 0.099, 0.377 and ≈ 1. Only 153 states (30.6%) satisfy both $\rho_H < 0.05$ and $|g| \le 3$, and the largest $|g|$ is 3.19. The fit is thus accurate near the anchors, but its HJB balance over the whole gate region is limited. This is the reason for the guard.

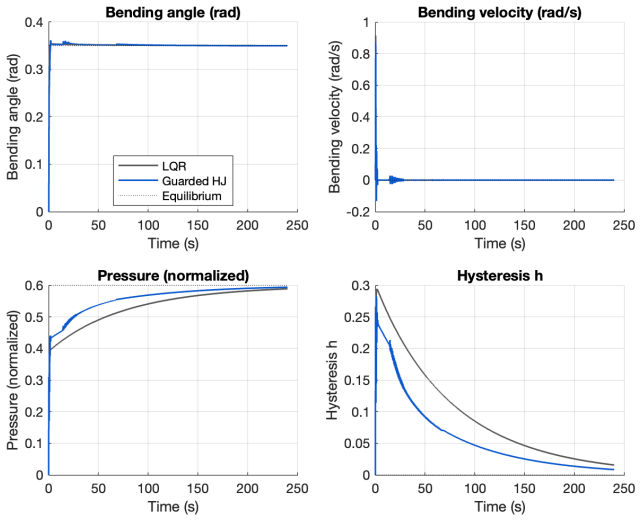


Fig. 4. Nominal regulation from rest ($B_d = 0$): angle, velocity, pressure and hysteresis under augmented LQR and guarded HJ.

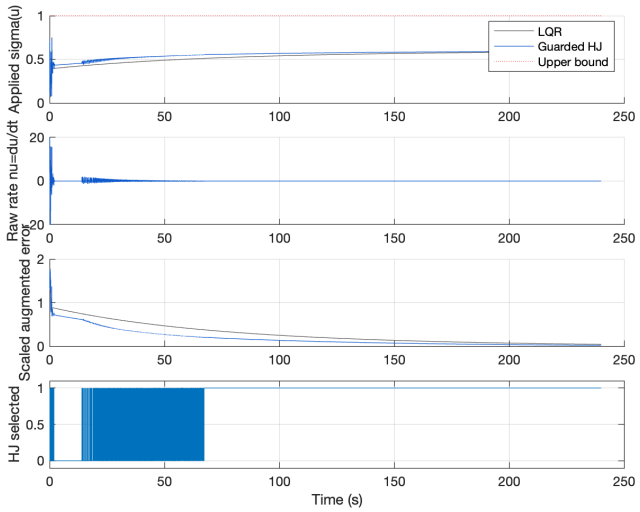


Fig. 5. Nominal applied command $\sigma(u)$, raw rate ν, scaled error $\lVert y\rVert$ and HJ-selection indicator (1: HJ rate applied).

5.3 Nominal regulation

Table 2 and Figs. 4–5 compare the two controllers. The HJ rate is applied on 18 382 of the 24 000 intervals (76.6%). The fallbacks comprise 10 domain rejections at the cold start and 5 608 residual rejections. The correction bound and the decrease guard (17) were never active. Guarded HJ lowers $\mathcal{J}_{240}$ by 20.8%, the tail RMSE by 34.5% and $\lVert e_z(240)\rVert$ by 46.0%. On the other hand, its full-horizon

Table 2. Nominal regulation from rest ($B_d = 0$, 240 s)

Metric	Aug. LQR	Guarded HJ
$\mathcal{J}_{240}$ (state + rate)	11.340	8.979
state part	11.071	7.573
rate part	0.269	1.406
Angle RMSE, full horizon (rad)	0.01174	0.01236
Angle RMSE, last 30 s (rad)	3.47×10^{-4}	2.27×10^{-4}
Settling time, ± 0.007 rad (s)	1.0	20.5
$\|e_z(240)\|$	0.0222	0.0120

RMSE is 5.3% higher, its rate cost is 5.23 times larger and it settles at 20.5 s instead of 1.0 s.

The gain does not come from the angle. The angle part of the state cost, $80 \cdot 240 \cdot \mathrm{RMSE}^2$, is 2.65 for LQR and 2.93 for HJ, whereas the remaining state cost falls from 8.43 to 4.64. HJ thus moves the slow pressure and hysteresis states (Remark 6) towards equilibrium faster; at 240 s the errors in p and h are -0.0058 and 0.0086, against -0.0108 and 0.0159 for LQR. Under HJ the raw command reaches -0.146, well outside $[0, 1]$, while the applied command stays above 0.070, as guaranteed by Proposition 2(c). The late settling coincides with the interval of roughly 14–68 s in which the selector alternates between HJ and LQR and the held rate oscillates (Fig. 5).

5.4 Periodic disturbance

The disturbance is a 0.4 Hz sinusoid w filtered by a lightly damped 0.7 Hz oscillator ($\xi_\eta = 0.15$), and the coupling B_d is swept. The peak torque reaches 0.172 at $B_d = 2$. Both controllers face identical forcing (Table 3). Guarded HJ has a lower cost and a lower tail RMSE in every case, but the margins are modest: cost reductions are 0.5–7.7%, and tail reductions range from 0.03% at $B_d = 0.15$ to 4.2% at $B_d = 2$. Because $\widehat{V}$ is designed for $d = 0$, these results are finite-horizon evidence and not a robustness certificate.

5.5 Discussion

Three observations guide the next design iteration. First, part of the trade-off in Table 2 is inherent to the cost (6). A policy that approximately minimizes $\mathcal{J}$ accepts a slightly larger angle transient in exchange for a faster reduction of the pressure and hysteresis errors, which LQR leaves to the 84-s mode of Remark 6. If angle settling has priority, Q_z can be reweighted, and the symbolic pipeline then regenerates $\overline{H}$, (10) and the canonical field without manual derivation. Second, the late band exits occur where the selector alternates between the two laws on consecutive samples (Fig. 5). A dwell time or a hysteresis band on the test $\rho_H \leq 0.05$ is a standard remedy for such chattering (Liberzon, 2003). It would, however, apply each law slightly beyond the states where it was validated. Third, the largest gap is geometric. Exact costates exist only for $\|y\| \leq 2.6 \times 10^{-4}$, whereas the controller acts up to $\|y\| = 1.6$, so over almost the whole operating region $\widehat{V}$ is shaped by the HJB residual alone. Extending the anchors along the characteristics, as in data-driven HJB methods (Kang and Wilcox, 2017; Nakamura-Zimmerer et al., 2021), requires no model change, because the anchors are generated by the same symbolic canonical field.

Table 3. Disturbance sweep: cost, tail angle RMSE (rad) and HJ use

B_d	$\mathcal{J}_{240}$ LQR	$\mathcal{J}_{240}$ HJ	$\Delta\mathcal{J}$	tail LQR	tail HJ	HJ use
0.15	12.98	11.98	-7.7%	0.01153	0.01153	71.5%
0.5	46.93	46.71	-0.5%	0.03846	0.03835	86.0%
1	165.20	161.78	-2.1%	0.07694	0.07608	92.9%
2	643.38	595.89	-7.4%	0.15405	0.14756	85.7%

6. CONCLUSION

Placing a smooth actuator saturation inside the drift of a rate-augmented model keeps the HJB problem input-affine. The optimal rate is then explicit, and the saturation slope appears automatically in the canonical equations. For a hysteretic soft pneumatic actuator, the stable manifold was computed with tight residuals, and a positive-definite value fit with a guarded selector lowered the regulation cost by 20.8% and the tail error by 34.5%. This came at the cost of more rate effort and slower angle settling. The limitations are explicit. The anchors reach only $\|y\| \approx 2.6 \times 10^{-4}$, whereas the cold start lies at 1.73, and only 30.6% of the held-out states pass the HJB test. Moreover, the rate bound and the switching are handled outside the HJB design. Future work will continue the manifold away from the origin with adaptive sampling, include the rate bound through a second saturation layer, treat the disturbance with a Hamilton–Jacobi–Isaacs formulation, and identify and test the model on a physical actuator.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work the authors used Claude (Anthropic) to survey the literature and draft and format the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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